Project Proposal Guide (Weeks 7–8)

This worksheet guides the team in developing the Capstone/Software Engineering Project Proposal. It integrates the refined problem statement, findings from the Review of Related Literature (RRL), and the proposed solution concept.

Team Information

Project Title: **Readlr**: A Gamified, Speech-Recognition Assisted Phonetics Learning Platform for Grade 1 Learners in the Philippines

Project Short Description (<20 words): A self-directed, gamified phonetics app that uses speech recognition to improve Grade 1 students' word pronunciation through interactive storytelling.

Team Code:2526-sem2-it332-27

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PART 1: Introduction (Approx. 300 words)

Reading proficiency, particularly accurate word pronunciation, stands as one of the most foundational competencies a child must develop during the early years of formal schooling. Yet in the Philippines, this foundation remains alarmingly fragile. According to the World Bank (2022), at least 90% of Filipino children aged 10 struggle to read or understand simple text, a crisis that originates as early as Grade 1, when children are first introduced to decoding written language. The Department of Education's Comprehensive Rapid Literacy Assessment further revealed that nearly half of students from Grades 1 to 3 were not reading at their respective grade levels at the end of School Year 2024–2025 (DepEd, 2025), underscoring the urgency of early phonetic and reading intervention.

A critical but often underexamined dimension of this literacy crisis is pronunciation. Librea et al. (2023) identified phonemic awareness, which is the capacity to absorb individual sounds of letters required for word recognition, as one of the most significant problems impeding children's reading comprehension in Philippine elementary schools. Compounding this, Padilla (2024) noted that some teachers who instruct beginning readers do not correctly model letter sounds, inadvertently teaching students inaccurate phonetic patterns from the earliest stages of literacy development. Canoy and Loquias (2022) further confirmed that mispronounced or substituted words among elementary students directly hindered reading fluency and comprehension performance.

Despite the availability of reading programs such as ECARP and Phil-IRI, these interventions remain largely paper-based, teacher-dependent, and insufficient in addressing pronunciation development at scale. The SEA-PLM (2024) assessment found that the Philippines' average reading score was statistically unchanged from 2019, with half of Filipino Grade 5 students still unable to understand simple written texts, which is evidence that existing approaches are not producing the systemic gains needed. The identified research gap lies in the absence of a locally contextualized, technology-driven tool that specifically targets phonetic pronunciation for beginning Grade 1 readers through interactive, child-centered means.

To address this gap, this project proposes a **gamified pronunciation and phonetics learning application** for Grade 1 Filipino learners. Using a storytelling-based approach, learners become the main character of an interactive adventure, encountering challenges that require correct pronunciation of phonetic sounds and vocabulary words to advance. The system employs speech recognition technology to deliver immediate and accurate pronunciation feedback (Sun, 2023), while game mechanics such as points, badges, and narrative rewards sustain motivation and engagement (Jayalath & Esichaikul, 2022). By merging immersive storytelling, gamification, and phonetics instruction, this application seeks to transform early pronunciation learning into an effective and enjoyable experience, one that builds speaking confidence, decoding skills, and reading readiness from the very start.

PART 2: Objectives

General Objectives:

General Objective 1: Improve Phonemic Awareness and Pronunciation Accuracy Through a Gamified, Voice-Driven Storytelling Application

- IImprove the phonemic awareness and pronunciation accuracy of Grade 1 learners by utilizing **Automated Speech Recognition (ASR) as the primary game controller**, targeting ≥80% pronunciation accuracy scores on CVC word assessments — consistent with DepEd's MELE benchmark for early grade reading readiness — ≥75% level completion rates across all game stages, and ≥85% successful ASR recognition of predefined phoneme inputs during gameplay sessions, based on constrained-vocabulary ASR performance standards reported in prior child-directed speech recognition studies (Shivakumar & Georgiou, 2020).

General Objective 2: Enhance Early Reading Readiness Through Voice-Activated Gameplay and Immediate Corrective Feedback

- Enhance early reading readiness among Grade 1 students by embedding an ASR-powered corrective feedback system into a narrative-based game, grounded in Mayer's Multimedia Learning Theory (2009) and Vygotsky's Zone of Proximal Development (1978), achieving ≥80% of learners demonstrating measurable improvement in phonemic decoding on post-test assessments, ≥75% of incorrect pronunciation attempts successfully corrected within the in-game feedback loop, and a reduction of phonetic mispronunciation errors by at least 30% from pre-intervention to post-intervention oral reading assessments.

General Objective 3: Promote Sustained Learner Engagement Through Gamification Mechanics, Reward Systems, and Self-Directed Interaction

- Promote consistent and measurable self-directed learning engagement — defined as unprompted, voluntary interaction with the application outside of teacher-facilitated sessions — among Grade 1 learners through gamification elements including level progression, voice-activated interactions, and a digital Sticker Book reward system anchored on Skinner's operant conditioning theory of positive reinforcement (1938), targeting ≥75% weekly active user retention, ≥70% daily reward milestone achievement, and ≥60% consistency in voluntary application use during the study period.

Specific Objectives

Specifically, this study seeks to:

For General Objective 1: Improve Phonemic Awareness and Pronunciation Accuracy Through a Gamified, Voice-Driven Storytelling Application

- Formulate a locally contextualized phonetics module by identifying and sequencing specific phonemes, CVC words, and early sight words aligned with DepEd's Most Essential Learning Competencies (MELCs) and the Marungko approach (Cagulada, 2018) — a government-endorsed localized phonetic reading strategy — with Level 1 targeting vowel sounds /a/, /e/, /i/, /o/, /u/ and Level 2 targeting consonant-vowel blending, covering all phoneme targets prescribed in the Grade 1 Filipino Language curriculum.
- Integrate an existing pre-trained ASR system (e.g., Google Speech-to-Text or OpenAI Whisper) configured for constrained pronunciation evaluation using a predefined vocabulary of phonemes and CVC words appropriate for Grade 1 learners, with grammar constraints and hotword boosting applied to improve recognition accuracy for child speech, targeting ≥85% recognition accuracy during controlled testing — in alignment with the accuracy threshold reported by Shivakumar and Georgiou (2020) for child-directed ASR systems.
- Implement a pronunciation confidence threshold scoring mechanism where inputs exceeding the threshold trigger correct-answer animations and story progression within ≤2 seconds of voice input detection, ensuring seamless and motivating gameplay response time.
- Design a centralized in-game progress dashboard that displays the learner's pronunciation scores, completed levels, and earned stickers in an organized and child-friendly interface, evaluated for usability using the ISO/IEC 25010 software quality model.
- Create ASR input validation prompts that detect silent, incomplete, or unclear voice entries and re-prompt the learner, targeting a reduction of unrecognized or unanswered pronunciation attempts by at least ≥70% during usability testing compared to baseline attempts without validation prompts.

For General Objective 2: Enhance Early Reading Readiness Through Voice-Activated Gameplay and Immediate Corrective Feedback

- Design a linear, level-based narrative progression map set within a contextualized story quest (e.g., finding missing letters to save a storybook kingdom), where each level directly maps to a specific phoneme or CVC word target from the DepEd-aligned phonetics module, structured in accordance with Vygotsky's scaffolding principle (1978) by progressing from simpler vowel sounds to more complex consonant blends.
- Implement voice-activated game mechanics where the player must correctly pronounce a letter sound or CVC word to execute game actions — such as saying the /m/ sound to make the character jump, or reading a word to unlock a treasure chest — with each voice interaction completing within ≤3 seconds of input, consistent with Nielsen's (1994) usability guideline for system response time in interactive applications.
- Embed an interactive corrective feedback loop for inputs below the confidence threshold, where an animated companion character demonstrates the correct mouth shape, plays the correct audio model of the target sound, and prompts the learner to retry, grounded in Mayer's Contiguity Principle (2009) — targeting ≥75% successful correction on the second attempt during testing.
- Conduct a quasi-experimental pre-test and post-test study using a modified Early Grade Reading Assessment (EGRA) or Phil-IRI oral reading instrument on a target group of Grade 1 learners to quantitatively measure improvements in phonemic decoding and pronunciation accuracy after the intervention period.
- Implement structured input validation that detects unanswered or silent game prompts and re-engages the learner with a visual and audio cue, targeting a reduction of abandoned game interactions by at least ≥70% during usability testing.

For General Objective 3: Promote Sustained Learner Engagement Through Gamification Mechanics, Reward Systems, and Self-Directed Interaction

- Implement a digital "Sticker Book" reward system where learners earn a collectible visual sticker representing local Filipino animals or story characters upon completing a level with high pronunciation accuracy, grounded in Skinner's positive reinforcement theory (1938), targeting ≥70% daily sticker milestone completion among active users during the study period.
- Design and evaluate a child-friendly game interface with responsive voice controls, age-appropriate visuals, and animated companions using the ISO/IEC 25010 software quality model across the dimensions of functional suitability, usability, reliability, and performance efficiency, with evaluation feedback gathered from at least five IT experts and three early childhood educators.
- Track and log individual learner session data — including session frequency, session duration, level completion rates, voice attempt counts per level, and reward milestone achievement — to operationalize and measure self-directed engagement as defined in General Objective 3 throughout the study period.
- Develop weekly in-game progress summaries displayed within the application that present completed levels, pronunciation accuracy trends, and collected stickers to reinforce intrinsic motivation and encourage continued voluntary participation,

consistent with Self-Determination Theory's emphasis on competence and autonomy as drivers of sustained engagement (Deci & Ryan, 1985)..

Research Questions

This study is guided by the following research questions:

1. To what extent does the use of the gamified pronunciation and phonetics learning application improve the word pronunciation accuracy of Grade 1 students, based on pre-test and post-test score comparisons?
2. Is there a statistically significant improvement in the phonemic awareness of Grade 1 students after completing the story-based application, as measured by pre-test and post-test assessment results?
3. How much improvement in oral reading fluency is observed among Grade 1 students before and after the self-directed use of the gamified application?
4. To what degree does the application reduce the frequency of phonetic mispronunciation errors among Grade 1 learners from pre-intervention to post-intervention oral reading assessments?
5. How engaged are Grade 1 students in self-directed learning throughout the application, as reflected by session frequency, level completion rates, and reward milestone achievement?

PART 3: Methods

The proposed solution is a **gamified pronunciation and phonetics learning application** designed for Grade 1 students in Philippine public elementary schools. The application operates as a self-directed learning tool, meaning students independently navigate and interact with the system without requiring teacher facilitation during use. The app employs a storytelling-based approach where the learner becomes the main character of an interactive adventure, progressing through story levels that present real phonetic challenges such as unlocking a door, helping a character, or finding an item, all of which require the student to correctly pronounce specific sounds, syllables, or vocabulary words in order to advance. The system uses built-in speech recognition technology to evaluate each learner's pronunciation in real time and provide immediate corrective feedback. To ensure technical feasibility and improve recognition accuracy among young learners, the ASR component operates within a constrained vocabulary environment that focuses only on predefined phonemes, syllables, CVC words, and Grade 1-level sight words rather than unrestricted conversational speech. Upon successfully completing each challenge, students earn points, badges, and narrative rewards that sustain motivation and encourage continued self-directed engagement. The primary users of this application are Grade 1 elementary learners aged 6 to 7 years old.

The development methodology will follow the Agile development framework, supporting iterative design, continuous feedback, and incremental feature deployment. Each sprint will focus on a specific functional component of the application, such as the speech recognition engine, the story level progression system, the phoneme challenge modules, and the reward and badge system, allowing for rapid testing, refinement, and quality assurance at each stage of development. Validation will involve usability testing with target-age learners and a pilot implementation conducted in a selected public elementary school environment.

To prove the application's educational effectiveness and directly address the defined objectives, student learning outcomes will be evaluated through a pre-test and post-test measurement design using the following measurable indicators:

- Word Pronunciation Accuracy. The percentage improvement in correct pronunciation scores between pre-test and post-test assessments, targeting a minimum gain of 30% among experimental group participants after the four-week intervention period.
- Phonemic Awareness. The percentage increase in students' ability to correctly identify, segment, and blend phonetic sounds, measured through a standardized phonemic awareness assessment administered before and after the intervention, targeting a minimum improvement of 25%.
- Oral Reading Fluency. The improvement in words read correctly per minute (WCPM) between pre-test and post-test oral reading fluency assessments, targeting a minimum gain of 25% among students who completed the intervention.
- Mispronunciation Error Reduction. The percentage decrease in the frequency of phonetic mispronunciation errors recorded during oral reading tasks from pre-intervention to post-intervention assessment, targeting a minimum reduction of 30%.
- Self-Directed Engagement. The session frequency, story level completion rate, and the percentage of students who achieved at least 70% of total in-app reward milestones throughout the four-week intervention period, as recorded by the application's built-in engagement logs.

PART 4: Expected System

Game Structure & Level Design

The application is structured as a single-player narrative campaign divided into **Stages (Chapters)** and **Levels**, mapped directly to the localized *Marungko* phonetics progression:

- **Stage 1: The Valley of Vowels (Basic Phonemes):** Comprises 5 introductory story levels. Each level isolates and targets a specific vowel sound (/a/, /e/, /i/, /o/, /u/). The player must accurately repeat the isolated phoneme to clear environmental obstacles.
- **Stage 2: The Blending Bridges (Consonant-Vowel Blending):** Focuses on combining consonant sounds with the vowels learned in Stage 1 (e.g., combining sounds to form basic syllables like *ma*, *ba*, *ta*).
- **Stage 3: The CVC Kingdom (Consonant-Vowel-Consonant Words):** Advanced narrative levels where challenges require the pronunciation of full CVC words (e.g., *cat*, *man*, *hat*) and early grade sight words to unlock items and progress the story.

The Micro-Gameplay Loop (Per Level):

1. **Hook:** An animated companion introduces a narrative conflict (e.g., *"The bridge is broken! Say the magic sound to fix it!"*).
2. **Model:** The companion models the correct pronunciation with on-screen text and high-quality audio.
3. **Action:** The child taps the microphone icon and speaks into the device.
4. **Evaluation:** If accurate, the character executes the action, the story advances, and a sticker is earned. If inaccurate, the corrective feedback loop triggers a scaffolded retry.

Minimum Viable Product (MVP) Features

1. Story-Based Adventure Interface

- An interactive narrative where the Grade 1 learner is the main character of a phonetics-driven adventure
- Age-appropriate illustrated scenes, characters, and story progressions designed for 6 to 7 year old learners
- Each chapter introduces a new phonetic concept aligned with Grade 1 literacy standards

2. Phonetic Challenge System

- Story-embedded challenges such as unlocking a door or rescuing a character that require correct pronunciation to advance
- Challenges progress in difficulty from simple vowel and consonant sounds to blended phonemes and multi-syllabic words

3. Speech Recognition and Feedback Engine

- Real-time evaluation of the learner's spoken pronunciation against a correct phonetic model
- Immediate corrective feedback after each attempt, with an audio model of the correct pronunciation provided for self-correction
- Multiple attempts are allowed per challenge before additional scaffolding support is provided

4. Reward and Progression System

- Students earn points, badges, and in-story rewards upon completing each phonetic challenge and story level
- Unlockable story chapters and collectible items motivate continued self-directed engagement

5. Phoneme and Vocabulary Bank

- A structured library of Grade 1 appropriate phonemes, words, and phrases sequenced by phonics difficulty
- All entries are accompanied by native-speaker quality audio models to establish correct pronunciation standards

6. Student Progress Dashboard

- A simple dashboard displaying completed levels, earned rewards, and overall pronunciation performance
- Progress is saved after every session to support continuity across multiple learning sessions

High-Level Workflow

1. Onboarding — The student is introduced to the story world, learns how to use the speech feature, and creates a basic profile to track progress across sessions

2. Story and Level Entry — The student enters the current story chapter, encounters a narrative problem, and is directed to the next phonetic challenge

3. Challenge Presentation — A visual prompt, written word or phoneme, and audio model are displayed, and the student is instructed to pronounce the target sound or word

4. Speech Recognition — The student speaks into the device, and the speech recognition engine evaluates the pronunciation against the correct phonetic model in real time

5. Feedback Delivery — Correct pronunciation advances the story with animations and rewards, while incorrect attempts receive gentle corrective feedback and another opportunity to try

6. Reward and Progression — Upon completing the challenge, the student earns points or badges and the story advances to the next scene

7. Session Summary — A session summary displays completed challenges and earned rewards, and the student is encouraged to return through a story preview of the next chapter.

PART 5: Discussion

Scope

The application targets Grade 1 students in Philippine public elementary schools, focusing on three core foundational literacy skills: word pronunciation accuracy, phonemic awareness, and oral reading fluency. It functions as a self-directed learning tool, requiring no teacher facilitation, and will be evaluated through a four-week quasi-experimental intervention involving 60 Grade 1 participants across experimental and control groups

Limitations

- Requires access to a compatible device with a functional microphone for speech recognition and internet connectivity

- Pilot testing limited to a selected public elementary school with 60 participants

- Speech recognition accuracy may vary due to background noise, microphone quality, and the phonetic diversity of young Filipino learners across different mother tongue languages

- The four-week intervention period may not capture long-term retention of pronunciation gains

- Self-directed engagement may vary among Grade 1 learners who are not yet accustomed to independent digital learning

Expected Contribution

This project directly addresses the research gap in locally contextualized, technology-driven pronunciation intervention for beginning Filipino readers. By combining speech recognition, gamification, and storytelling-based phonetics instruction, the application provides Grade 1 learners with an engaging and measurable self-directed learning experience. It produces empirical evidence on the effectiveness of gamified phonetic tools in improving pronunciation accuracy, phonemic awareness, and oral reading fluency, contributing to the growing body of educational technology research in the Philippine early literacy context and offering a scalable, replicable model for future digital reading interventions aligned with the K to 12 curriculum.

PART 6: Traceability Matrix

RRL Finding/Theme	Identified Gap	Research Question	Proposed Function
At least 90% of Filipino children aged 10 struggle to read and understand simple text, with the crisis rooted as early as Grade 1 (World Bank, 2022)	No locally developed, technology-driven tool that specifically targets word pronunciation for Grade 1 beginning readers	To what extent does the application improve the word pronunciation accuracy of Grade 1 students based on pre-test and post-test scores?	Phonetic Challenge System with story-embedded pronunciation tasks that require correct word and sound production to advance
Phonemic awareness, the ability to identify and reproduce individual letter sounds, is one of the most significant barriers to early reading development among Filipino elementary	Absence of a structured, interactive phoneme practice tool designed specifically for self-directed use by Grade 1 learners	Is there a statistically significant improvement in phonemic awareness scores before and after use of the application?	Phoneme and Vocabulary Bank with sequenced phoneme sets covering vowels, consonants, blends, and digraphs with native-speaker audio models

learners (Librea et al., 2023)			
Some teachers instruct beginning readers with incorrect letter sounds, compounding early pronunciation deficiencies in the classroom (Padilla, 2024)	No self-directed tool that allows Grade 1 learners to access correct phonetic models and practice independently outside of teacher instruction	How much improvement in oral reading fluency is observed among Grade 1 students before and after the intervention?	Speech Recognition and Feedback Engine that evaluates spoken pronunciation in real time and delivers immediate corrective feedback with correct audio models
Mispronounced and substituted words during oral reading directly hinder reading fluency and comprehension among Filipino elementary students (Canoy & Loquias, 2022)	Lack of an automated, real-time pronunciation evaluation tool accessible to young learners without requiring human assessment	To what degree does the application reduce the frequency of phonetic mispronunciation errors from pre-intervention to post-intervention assessments?	Speech Recognition Engine that detects pronunciation errors per attempt and provides scaffolded corrective feedback across multiple tries
The Philippines' average reading score remained statistically unchanged from 2019 to 2024, with half of Grade 5 students still unable to understand simple texts, indicating that existing interventions are insufficient (SEA-PLM, 2024)	Existing reading programs such as Phil-IRI and ECARP are paper-based, teacher-dependent, and do not address pronunciation development through interactive or digital means n	Is there a statistically significant difference in post-test scores between the experimental group using the app and the control group using regular classroom instruction?	Quasi-Experimental Pre-test and Post-test Design with experimental and control groups to measure and compare learning gains produced by the application
Game-like features and progress tracking in technology-based tools offer a more motivating and engaging alternative	No gamified, self-directed pronunciation application designed specifically for the phonetic	How engaged are Grade 1 students throughout the application as reflected by session frequency, level completion rates,	Reward and Progression System featuring points, badges, unlockable story chapters, and a student progress dashboard to

to traditional classroom methods for young learners (Jayalath & Esichaikul, 2022)	development needs of Grade 1 Filipino learners	and reward milestone achievement?	sustain self-directed engagement

PART 7: References

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